Control of Mechanical Engineering Systems
University of Florida
Mechanical and Aerospace Engineering

Final Project
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Problem 1:

$$f_1 = x_7$$
$$f_2 = x_8$$
$$f_3 = x_9$$
$$f_4 = x_{10}$$
$$f_5 = x_{11}$$
$$f_6 = x_{12}$$
$$f_7 = -\frac{1}{M+m}[sin(\phi)sin(\psi) + cos(\phi)cos(\psi)sin(\theta)]u_1$$
$$f_8 = -\frac{1}{M+m}[cos(\phi)sin(\psi)sin(\theta) - cos(\psi)sin(\phi)]u_1$$
$$f_9 = g - \frac{cos(\phi)cos(\theta)}{M+m}u_1$$
$$f_{10} = \frac{I_y - I_z}{I_x}(\dot{\theta})(\dot{\psi}) + \frac{u_2}{I_x}$$
$$f_{11} = \frac{I_z - I_x}{I_y}(\dot{\phi})(\dot{\psi}) + \frac{u_3}{I_y}$$
$$f_{12} = \frac{I_x - I_y}{I_z}(\dot{\phi})(\dot{\theta})$$

Problem 2:

- $\ddot{x} \approx f(x_o, u_o) + \frac{\partial f}{\partial x}|_{(x_o,u_o)}\big(x(t) - x_o(t)\big) + \frac{\partial f}{\partial u}|_{(x_o,u_o)}\big(u(t) - u_o(t)\big)$

$$\ddot{x} \approx -\frac{1}{M+m}[\sin(\phi)\sin(\psi) + \cos(\phi)\cos(\psi)\sin(\theta)](M+m)g + 0$$
$$-\frac{1}{M+m}[\sin(\phi)\sin(\psi) + \cos(\phi)\cos(\psi)\sin(\theta)]\big(u_1(t) - u_{1o}(t)\big)$$
$$\approx -g[0 + 1\sin(\theta)] - \tfrac{1}{M+m}[0 + 1\sin(\theta)]\tilde{u}_1(t)$$
$$\approx -g[x_5] - \tfrac{1}{M+m}[x_5]\tilde{u}_1(\mathrm{t})$$
$$\approx -gx_5 - 0$$
$$\ddot{x}_7 \approx -gx_5$$

- $\ddot{y} \approx f(x_o, u_o) + \frac{\partial f}{\partial x}|_{(x_o,u_o)}\big(x(t) - x_o(t)\big) + \frac{\partial f}{\partial u}|_{(x_o,u_o)}\big(u(t) - u_o(t)\big)$

$$\ddot{y} \approx -\frac{1}{M+m}[cos(\phi)sin(\psi)sin(\theta) - cos(\psi)sin(\phi)](M+m)g + 0$$
$$-\frac{1}{M+m}[cos(\phi)sin(\psi)sin(\theta) - cos(\psi)sin(\phi)]\big(u_1(t) - u_{1o}(t)\big)$$
$$\approx -g[0 - 1\sin(\phi)] - \tfrac{1}{M+m}[0 - 1\sin(\phi)]\tilde{u}_1(t)$$
$$\approx -g[-x_4] - \tfrac{1}{M+m}[-x_4]\tilde{u}_1(t)$$
$$\approx gx_4 - 0$$
$$\ddot{\tilde{x}}_8 \approx gx_4$$

- $\ddot{\tilde{z}} \approx f(x_o, u_o) + \frac{\partial f}{\partial x}|_{(x_o,u_o)}\big(x(t) - x_o(t)\big) + \frac{\partial f}{\partial u}|_{(x_o,u_o)}\big(u(t) - u_o(t)\big)$

$$\ddot{\tilde{z}} \approx g - \frac{cos(\phi)cos(\theta)}{M+m}(M+m)g + 0 - \frac{cos(\phi)cos(\theta)}{M+m}\big(u_1(t) - u_{1o}(t)\big)$$
$$\approx g(1 - cos(\phi)cos(\theta)) - \frac{1}{M+m}\tilde{u}_1(t)$$
$$\approx g(1-1) - \frac{1}{M+m}\tilde{u}_1(t)$$
$$\ddot{\tilde{x}}_9 \approx -\frac{1}{M+m}\tilde{u}_1(t)$$

- $\ddot{\tilde{\phi}} \approx f(x_o, u_o) + \frac{\partial f}{\partial x}|_{(x_o,u_o)}\big(x(t) - x_o(t)\big) + \frac{\partial f}{\partial u}|_{(x_o,u_o)}\big(u(t) - u_o(t)\big)$

$$\ddot{\tilde{\phi}} \approx \frac{I_y - I_z}{I_x}(\dot{\theta})(\dot{\psi}) + \frac{u_{2o}(t)}{I_x} + 0 + \frac{1}{I_x}\big(u_2(t) - u_{2o}(t)\big)$$
$$\approx \frac{I_y - I_z}{I_x}(0) + \frac{0}{I_x} + 0 + \frac{1}{I_x}\tilde{u}_2(t)$$
$$\ddot{\tilde{x}}_{10} = \frac{1}{I_x}\tilde{u}_2(t)$$

- $\ddot{\tilde{\theta}} \approx f(x_o, u_o) + \frac{\partial f}{\partial x}|_{(x_o,u_o)}\big(x(t) - x_o(t)\big) + \frac{\partial f}{\partial u}|_{(x_o,u_o)}\big(u(t) - u_o(t)\big)$

$$\ddot{\tilde{\theta}} \approx \frac{I_z - I_x}{I_y}(\dot{\phi})(\dot{\psi}) + \frac{u_{3o}(t)}{I_y} + 0 + \frac{1}{I_y}\big(u_3(t) - u_{3o}(t)\big)$$
$$\approx \frac{I_z - I_x}{I_y}(0) + \frac{0}{I_y} + 0 + \frac{1}{I_y}\tilde{u}_3(t)$$
$$\ddot{\tilde{x}}_{11} \approx \frac{1}{I_y}\tilde{u}_3(t)$$

- $\ddot{\psi} \approx f(x_o, u_o) + \frac{\partial f}{\partial x}|_{(x_o,u_o)}\big(x(t) - x_o(t)\big) + \frac{\partial f}{\partial u}|_{(x_o,u_o)}\big(u(t) - u_o(t)\big)$

  $$\ddot{\psi} \approx \frac{I_x - I_y}{I_z}(\dot{\phi})(\dot{\theta}) + 0 + 0$$

  $$\approx \frac{I_x - I_y}{I_z}(0) + 0 + 0$$

  $\ddot{\tilde{x}}_{12} = 0$

Written in linear state space equation form:

$$\begin{pmatrix} \dot{\tilde{x}}_1 \\ \dot{\tilde{x}}_2 \\ \dot{\tilde{x}}_3 \\ \dot{\tilde{x}}_4 \\ \dot{\tilde{x}}_5 \\ \dot{\tilde{x}}_6 \\ \dot{\tilde{x}}_7 \\ \dot{\tilde{x}}_8 \\ \dot{\tilde{x}}_9 \\ \dot{\tilde{x}}_{10} \\ \dot{\tilde{x}}_{11} \\ \dot{\tilde{x}}_{12} \end{pmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -g & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & g & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \tilde{x}_4 \\ \tilde{x}_5 \\ \tilde{x}_6 \\ \tilde{x}_7 \\ \tilde{x}_8 \\ \tilde{x}_9 \\ \tilde{x}_{10} \\ \tilde{x}_{11} \\ \tilde{x}_{12} \end{pmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{-1}{M+m} & 0 & 0 \\ 0 & \frac{1}{I_x} & 0 \\ 0 & 0 & \frac{1}{I_y} \\ 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} \tilde{u}_1(t) \\ \tilde{u}_2(t) \\ \tilde{u}_3(t) \end{pmatrix}$$

$$\begin{pmatrix} \tilde{y}_1 \\ \tilde{y}_2 \\ \tilde{y}_3 \\ \tilde{y}_4 \\ \tilde{y}_5 \\ \tilde{y}_6 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \tilde{x}_4 \\ \tilde{x}_5 \\ \tilde{x}_6 \\ \tilde{x}_7 \\ \tilde{x}_8 \\ \tilde{x}_9 \\ \tilde{x}_{10} \\ \tilde{x}_{11} \\ \tilde{x}_{12} \end{pmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} \tilde{u}_1(t) \\ \tilde{u}_2(t) \\ \tilde{u}_3(t) \end{pmatrix}$$

Problem 3: (Code is written in param.m)

- *MATLAB Results:*

- From input 1 to output...
  1: 0
  2: 0
  3: $-\frac{0.0006557}{s^2}$
  4: 0
  5: 0
  6: 0

- From input 2 to output...
  1: 0
  2: $-\frac{1.752e-5}{s^4}$
  3: 0
  4: $\frac{4.72e-6}{s^2}$
  5: 0
  6: 0

- From input 3 to output...
  1: $-\frac{1.753e-5}{s^4}$
  2: 0
  3: 0
  4: 0
  5: $\frac{4.732e-6}{s^2}$
  6: 0

- *Answers:*
  - The system is not BIBO stable for any input to any output. As can be seen from the result, all transfer functions have either 2 or 4 poles at the origin, which make the system not BIBO stable.
  - The sign of the transfer function from u_1 to y_3 is negative because we are navigating in the coordinate frame where z axis is pointed downwards. Input 1 is the deviation in thrust applied, and by increasing thrust, we are telling the sky crane to move upwards (moving in the negative z direction).
  - For an impulse response:

$$h(t) = \mathcal{L}^{-1}\{H(s) \cdot 1\} = \mathcal{L}^{-1}\left\{-\frac{0.0006557}{s^2}\right\}$$
$$h(t) = -0.0006557t$$

Because the result has variable t, it indicates that the crane altitude/height will continue to increase as time increases and eventually drift to infinity. This makes sense because the input is a deviation input. An impulse represents a momentary burst of thrust above the equilibrium needed for hover. This burst changes a step change in the crane's vertical velocity. Since there is no restoring force in the open-loop system, the crane maintains this new velocity, causing the altitude deviation to increase linearly over time.

Problem 4: (Code written in FP_P4.m)

- For $K_P = 10$ and $K_I = 1$

$$1 + C(s)P(s) = (K_D s + 10 + \frac{1}{s})\left(\frac{-0.00066557}{s^2}\right)$$

$$1 + C(s)P(s) = K_D\left(\frac{0.00066557s^2}{s^3 + 0.006557s + 0.00066557}\right)$$

$$M(s) = \frac{0.00066557s^2}{s^3 + 0.006557s + 0.00066557}$$

Plot:

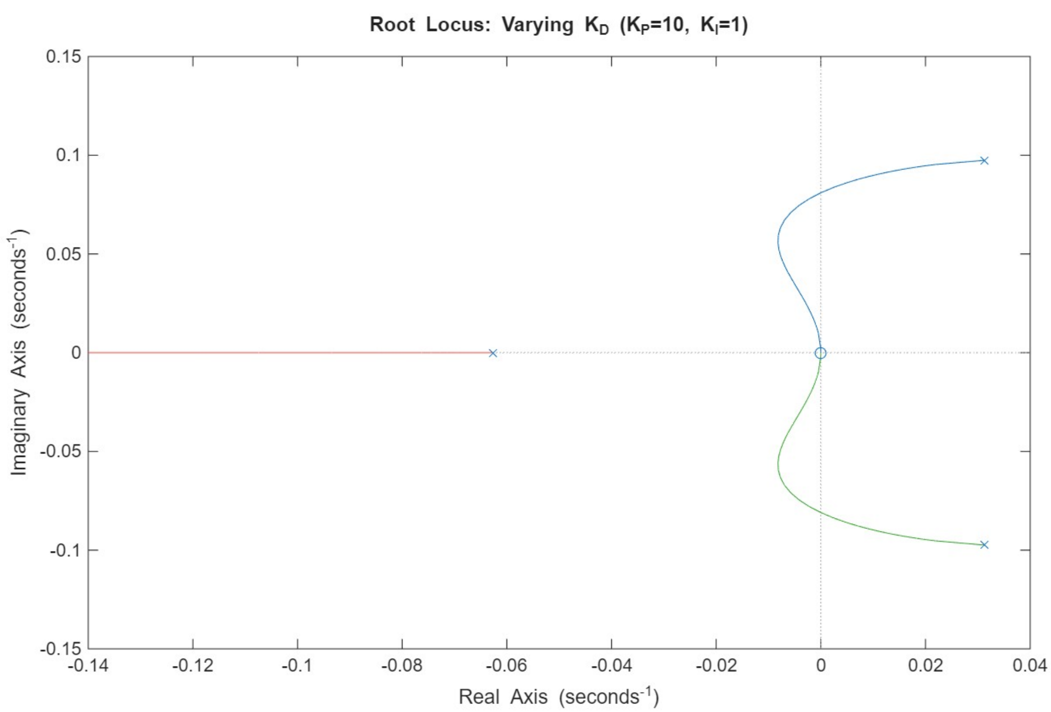

Figure 1: *Root Locus With varying K_D*

Based on the plot, the system is initially unstable for the chosen K_P and K_I. However, as K_D increases, the system eventually become stable as the two initial poles on the RHP will cross over to the SLHP for some time before ending at the zero at the origin. Additionally, because there are two complex poles, there will be oscillation in the step response. Lastly, as K_D increases, the system will respond slower. This is because the dominant pole is moving towards the origin.

- For $K_D = 1$ and $K_I = 1$

$$1 + C(s)P(s) = (s + K_P + \frac{1}{s})\left(\frac{-0.00066557}{s^2}\right)$$
$$1 + C(s)P(s) = K_P\left(\frac{0.00066557s}{s^3 + 0.0006557s^2 + 0.00066557}\right)$$
$$M(s) = \frac{0.00066557s}{s^3 + 0.0006557s^2 + 0.00066557}$$

Plot:

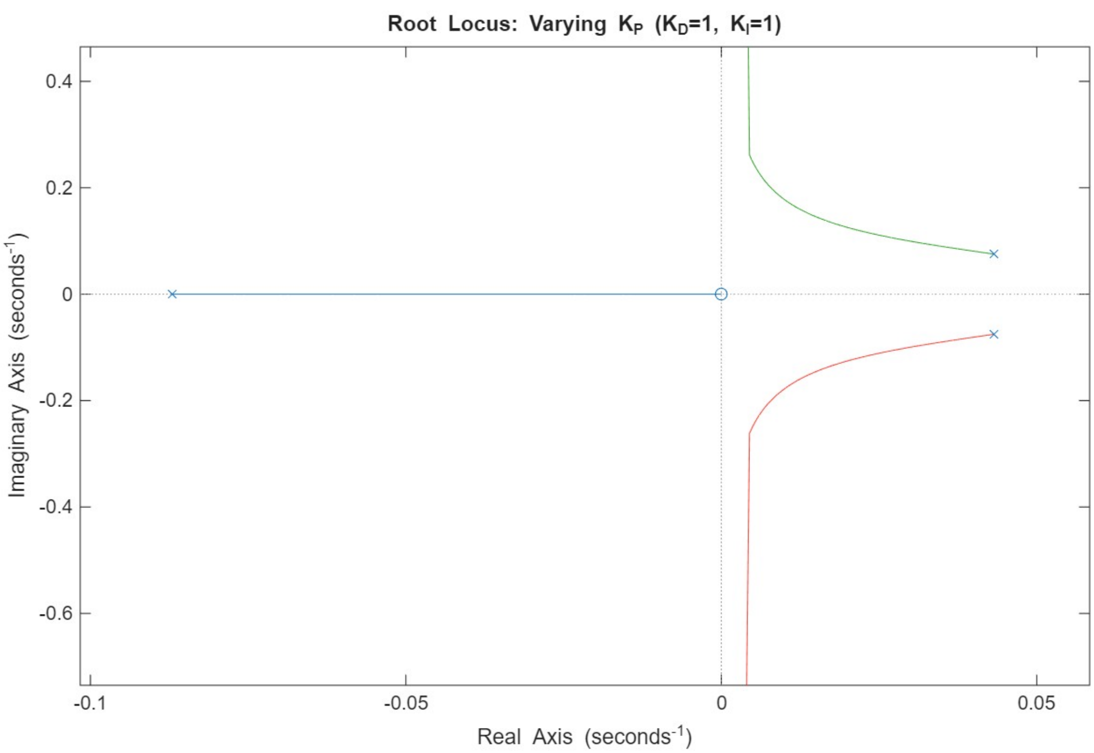

Figure 2: *Root Locus With varying K_P*

This system can never be stable, even if K_P is increased. As shown on the plot, two of the poles (originally on the RHP) will never cross the imaginary axis and lie on the SLHP. There will be oscillations because the t wo poles on the RHP are complex poles. As K_P increases, those two poles will continue to infinity and -infinity parallel to the imaginary axis. In this system, the system response will also become slower with increasing K_P, because the dominant pole is moving towards a zero at the origin.

- For $K_D = 10$ and $K_P = 10$

$$1 + C(s)P(s) = (10s + 10 + \frac{K_I}{s})\left(\frac{-0.00066557}{s^2}\right)$$
$$1 + C(s)P(s) = K_P\left(\frac{0.00066557s}{s^3 + 0.006557s^2 + 0.0066557s}\right)$$
$$M(s) = \frac{0.00066557s}{s^3 + 0.006557s^2 + 0.0066557s}$$

Plot:

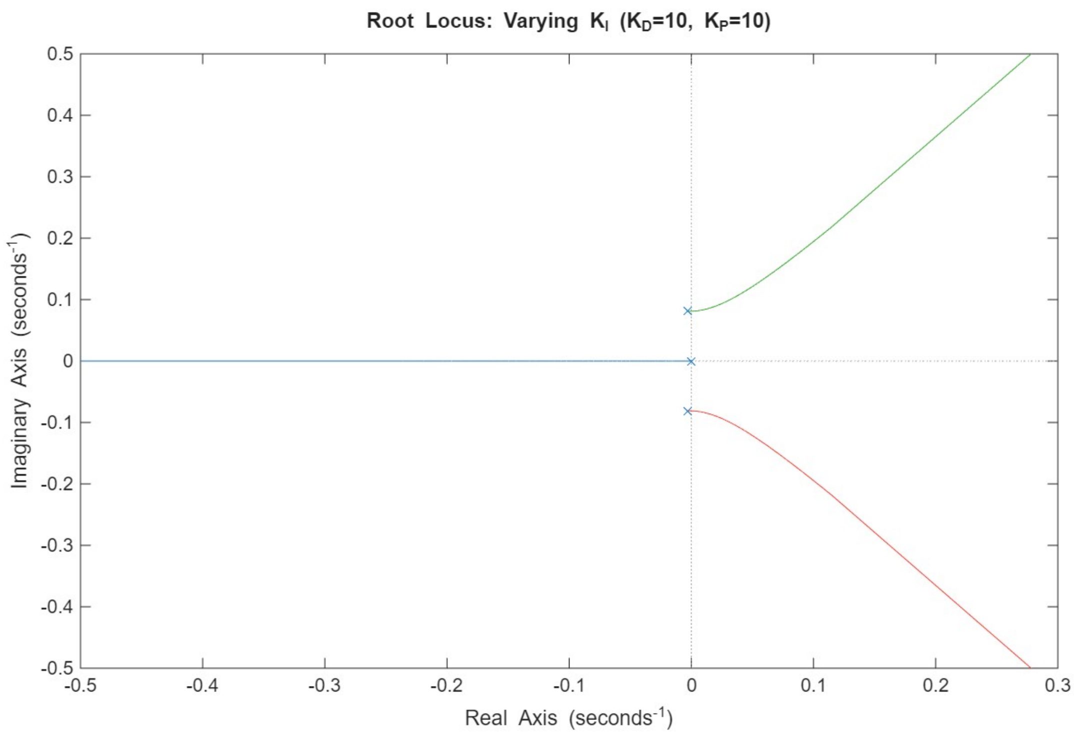

Figure 3: *Root Locus With varying K_I*

This system starts off being stable with very low gain K_I. However, as the gain is increased, the system becomes unstable as two poles move towards the RHP. There are also oscillations in this system, because two of the poles are always complex. As for the response time, As KI increases, the complex conjugate poles move towards infinity on the RHP, increasing the natural frequency. As the natural frequency increases, the rise time drops and the response time becomes faster.

Problem 5: (Code written in FP_P5.m)

To meet all the requirements (no overshoot, rise time < 15s, and no steady state error) I Initially, chose my PID gains to be as follows: $K_D = 9000$, $K_P = 400$, and $K_I = 0.5$. The plot of step response of the chosen controller gains to a sudden change in height from 0 to 10 meters is as shown in Figure 4.

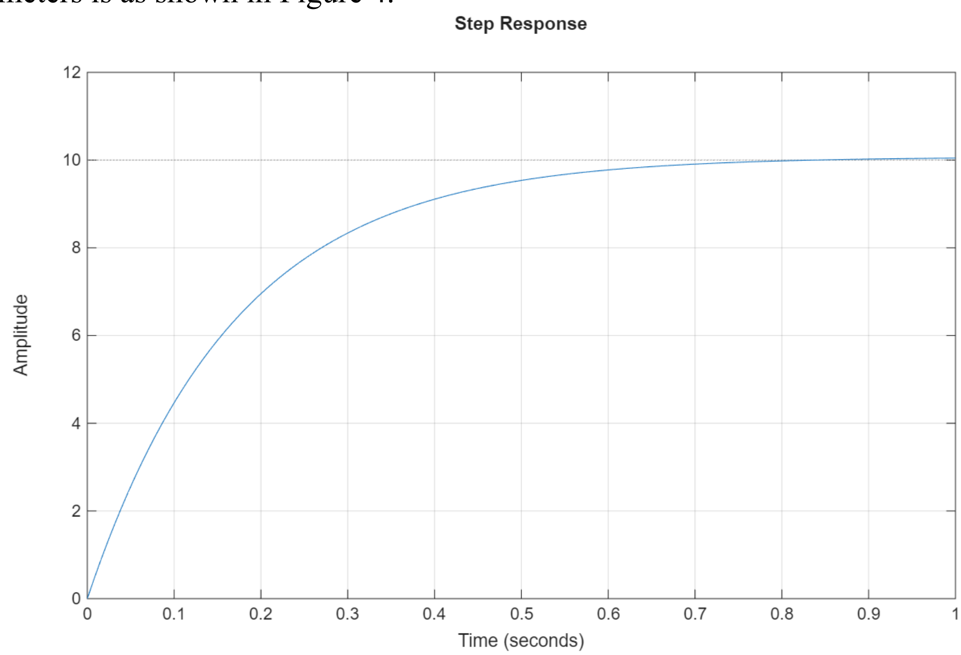

Figure 4: *PID Controller First Attempt Result*

Based on the plot, the chosen controller has a rise time of 0.3642 seconds, 0 percent overshoot, and a steady state error of -0.043, which satisfies all the requirements. However, when the chosen controller was used for problem 7, the low K_P does not work well in tracking the given height references. To have better reference tracking in problem 7 simulation, the K_P must be increased. An increase in K_P, however leads to an overshoot in problem 5. It is almost impossible to simultaneously satisfy the requirements of both problems 5 and 7. I tried different gains until arriving at $K_D = 9000$, $K_P = 4000$, and $K_I = 0.5$. The step response plot of the chosen gain is shown in Figure 5.

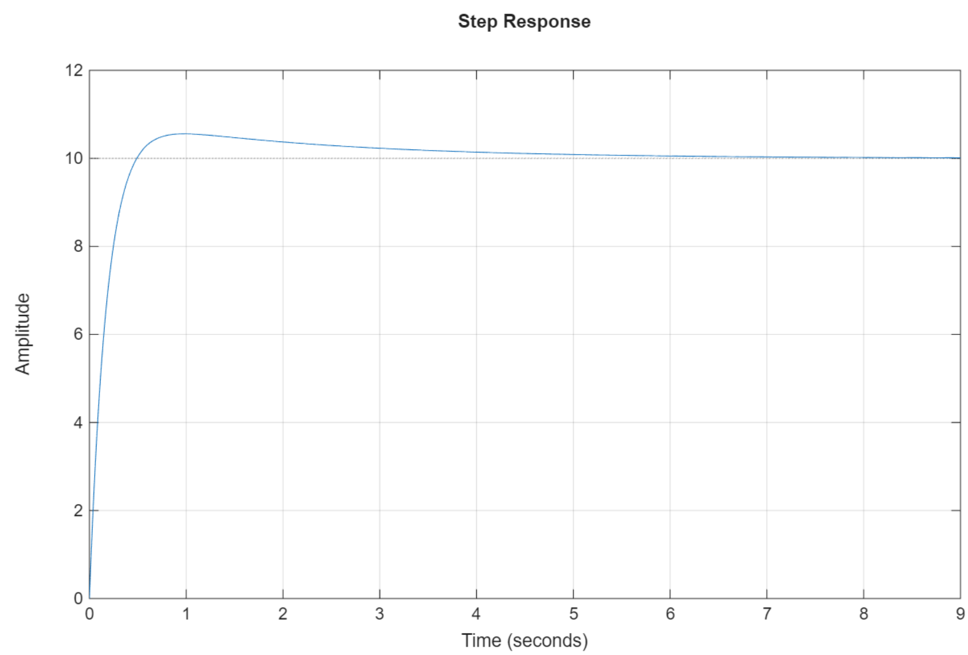

Figure 5: *PID Controller Final Attempt Result*

According to figure 5, while the rise time and steady state error requirements are satisfied, there is a slight overshoot at around 0.5 seconds. However, this is not of major concern because the controller works well with no overshoot on problem 7. As Dr. brooks stated, if it satisfies problem 7, it also satisfies problem 5. So, I decided to go with the second set of gain ($\underline{K_D = 9000, K_P = 4000, \text{and } K_I = 0.5}$) rather than the first set of values.

Problem 6: (Code written in FP_P6.m) (The Bode Plot File is called: FP_P6_Design.mat)

- Requirement 1:

$$|L(j\omega)| > \frac{1}{\varepsilon_r}$$
$$|L(j1)| > \frac{1}{0.01}$$
$$|L(j1)| > 100$$
$$20log_{10}(100) = 40 \text{ dB}$$

Therefore for $\omega = 1$ rad/s the magnitude must be above 40 dB. And unwanted region is shaded in orange on Figure 6.

- Requirement 2:

Since the given transfer function has two poles at the origin, there is already no steady state error to a step reference. But I also double check with the plot on controlSystemDesigner.

- Requirement 3:

$$|L(j\omega)| < \varepsilon_n$$
$$|L(j100)| < \frac{1}{100}$$
$$|L(j100)| < 0.01$$
$$20log_{10}(0.01) = -40 \text{ dB}$$

Therefore for $\omega = 100$ rad/s the magnitude must be below -40 dB. And unwanted region is shaded in orange on Figure 6.

- Requirement 4:

To ensure closed loop stability, gain margin must be positive and phase margin must be positive.

In order to satisfy all given requirements, I started by plotting the given transfer function for pitch $P(s) = \frac{4.732e{-6}}{s^2}$ on MATLAB's controlSystemDesigner tool. Then I started by setting the constraints on the plot, using Design Requirements, based on requirement 1 and 3. Then I set the gain to 100,000 to account for the 4.732e-6 term in $P(s)$. Later I started by placing 3 poles before 1 rad/s to bring the phase up from -180 degrees and bring the magnitude up to satisfy requirement 1. However, with three real zeroes added, the magnitude is still below 40 dB at 1 rad/s. So, I added a pair of complex poles with a natural frequency of 1 rad/s and a damping ratio of 0.003 to create a peak in magnitude that sits just right above 40 dB. However, because more zeros were added, requirement 3 is still not satisfied. In order to be just below -40 dB at 100 rad/s, 3 more real poles were added to bring the magnitude down. Lastly, I checked whether gain margin and phase margin were positive and that the steady state error to a step input is zero based on the plot on controlSystemDesigner tool. The final controller transfer function is:

$$C(s) = (8.3333 \times 10^8)\frac{(s+0.1)(s+0.3)(s+0.4)}{(s+2)(s+50)(s+50)(s^2+0.006s+1)}$$

The magnitude and phase plot of $L(s)$ is plotted on Figure 6. Step response is plotted on Figure7.

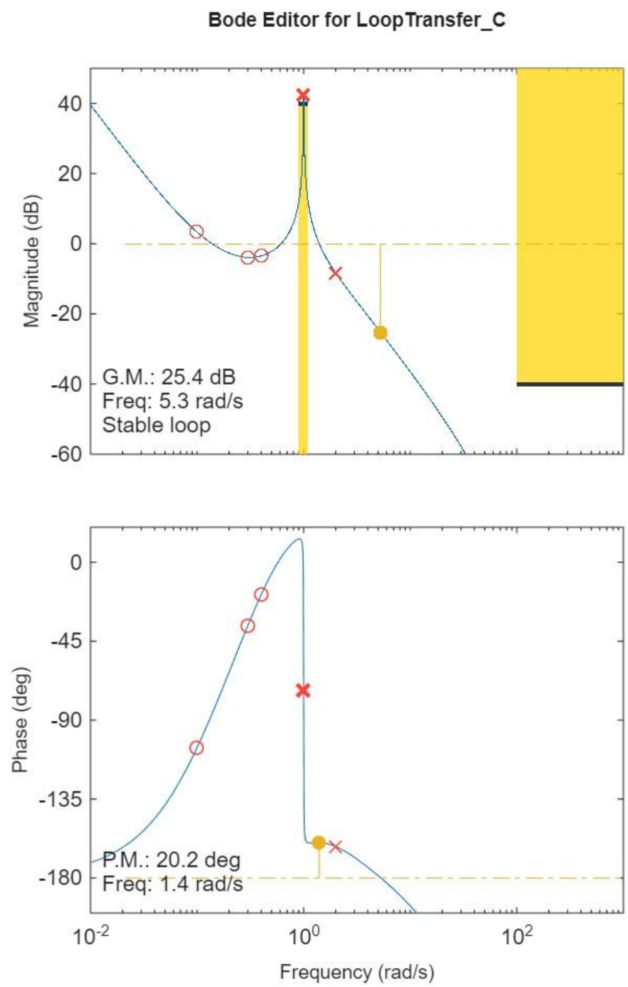

Figure 6: *Phase and Magnitude Plot*

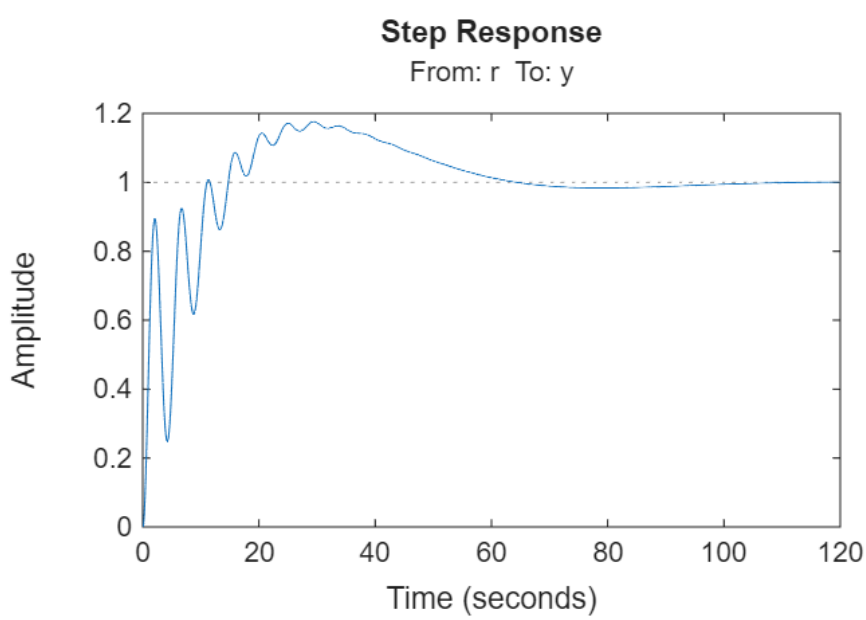

Figure 7: *Step Response Plot*

As can be seen in Figure 6, magnitude peaks at 1 rad/s above 40 dB, satisfying requirement 1. The magnitude is also below -40 dB at 100 rad/s, satisfying requirement 3. The second requirement is also satisfied as shown in Figure 7. Although there were oscillations in the response, it eventually dampened down to the steady state. Lastly the gain margin of 25.4 dB and phase margin of 20.2 degrees also suggest that the fourth requirement is satisfied.

There were other controllers $C(s)$, that I tried that satisfied all the requirements while also having less oscillations in the step response plot. However, those usually resulted in higher DC gain that when input into the Simulink model, drives the crane altitude to infinity. Additionally, other iterations also lead to worse reference tracking of pitch angles as well as more saturation in the control command, causing the controller to sometimes behave like a Bang-Bang controller.

Problem 7: (Code written in linsim_script.m) (Simulink Model in linsim.slx)

- Simulink Model:

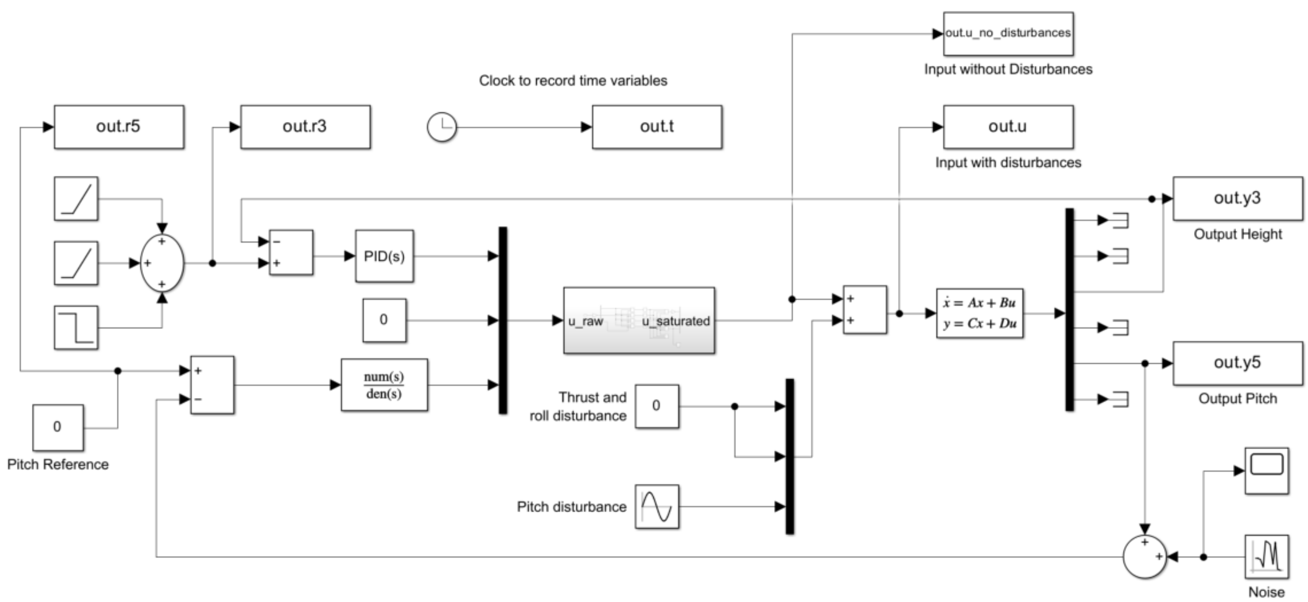

Figure 8: *Linear Simulation Simulink Model*

- Plots:

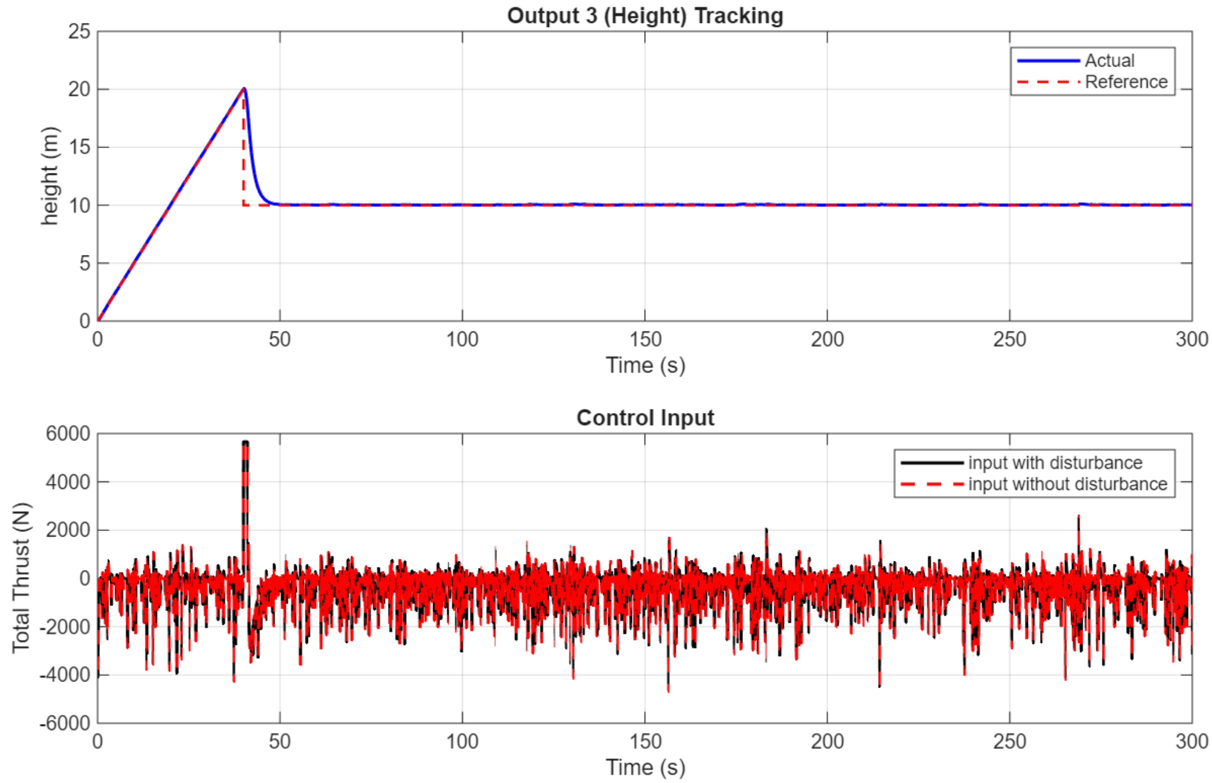

Figure 9: *$\tilde{y}_3$ (Height) and its reference vs time and $\tilde{u}_1$ (Total Thrust) vs time.*

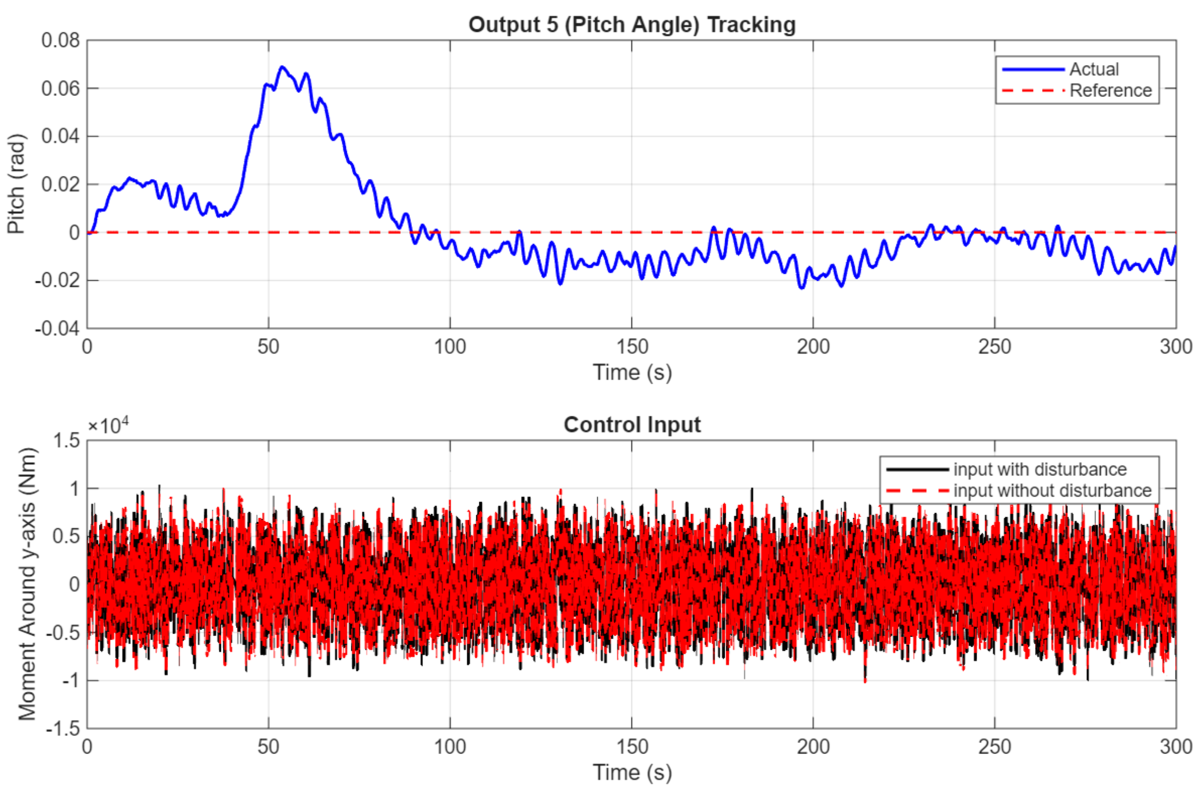

Figure 10: *$\tilde{y}_5$ (Pitch) and its reference vs time and $\tilde{u}_3$ (Total Moment about y-axis) vs time.*

According to Figure 10, my chosen controller from problem 6 seems to have more oscillations in pitch control than Dr Brook's plot. However, I chose to go with my controller, because it is better at reference tracking than Dr Brook's. I tried removing oscillations by reducing the damping coefficient in my complex poles, but that would affect the gain, which make the height reference tracking become worse as well. So, I decided to stick to the stated controller.

Problem 8: (Code written in nonlinsim_script.m) (Simulink Model in nonlinsim.slx)

- Simulink Model:

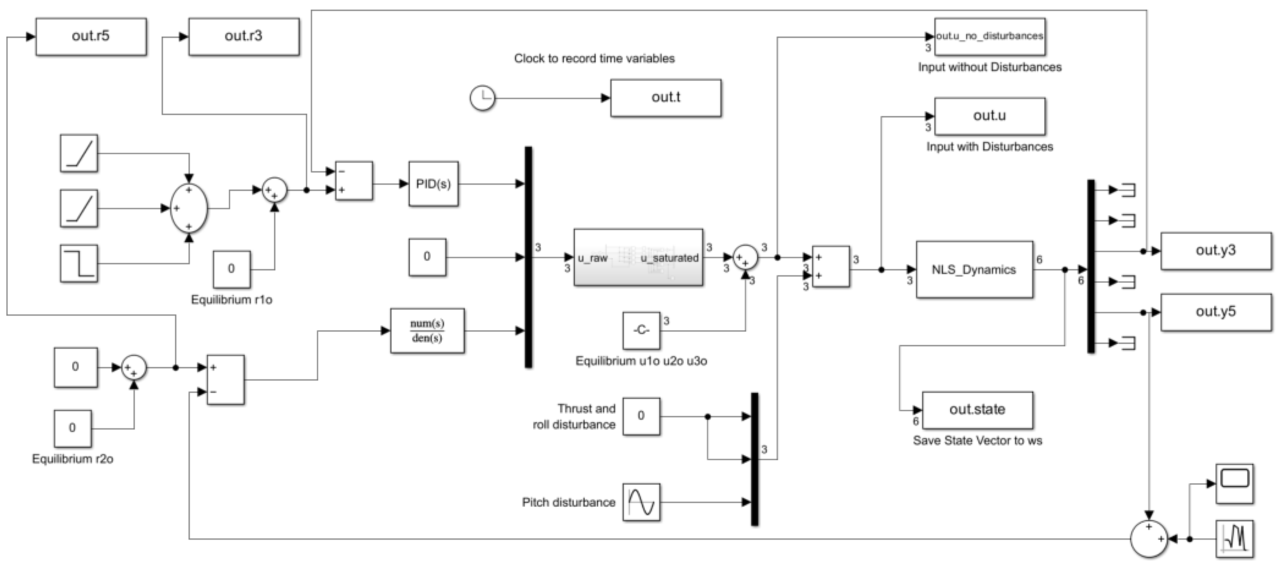

***Figure 11:** Non-Linear Simulation Simulink Model*

- Plots:

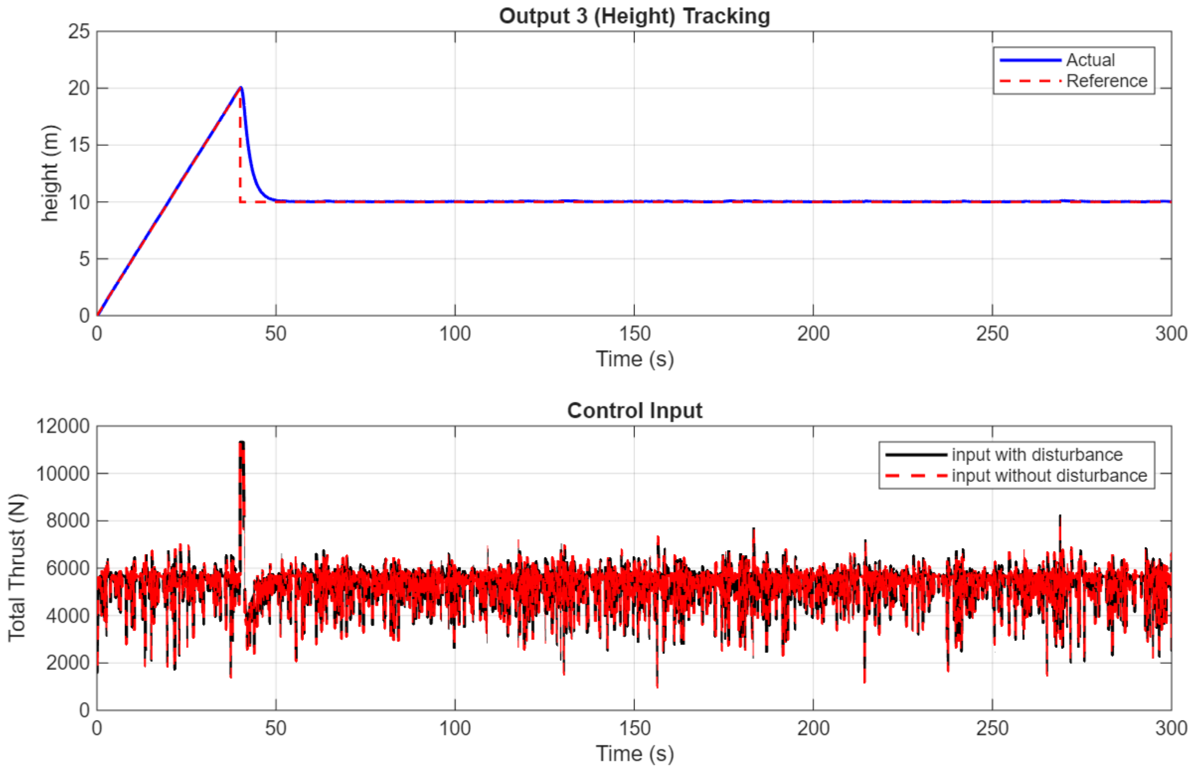

Figure 12: *$\tilde{y}_3$ (Height) and its reference vs time and $\tilde{u}_1$ (Total Thrust) vs time.*

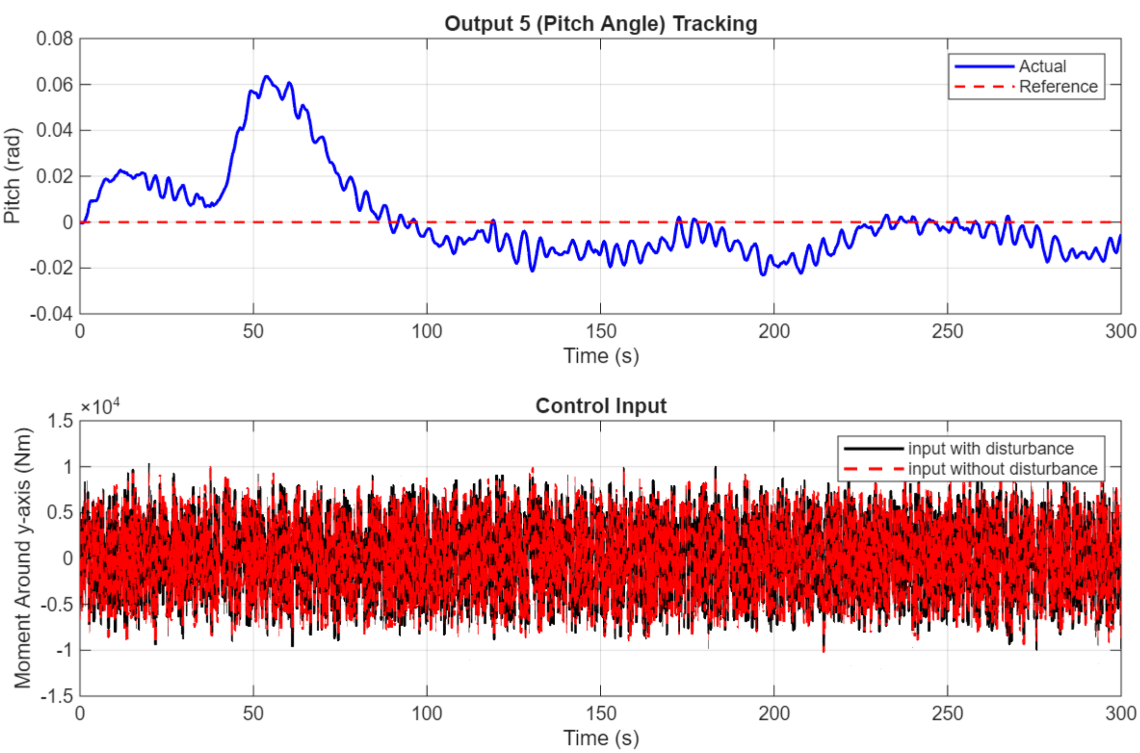

Figure 13: *$\tilde{y}_5$ (Pitch) and its reference vs time and $\tilde{u}_3$ (Total Moment about y-axis) vs time.*

Problem 9: (Code written in FP_P9.m) (Simulink Model in nonlinsim_bonus.slx)

- My chosen Poles:

$$[-0.2, -0.3, -0.4, -0.5, -0.6, -0.7, -0.8, -0.9, -1.0, -1.1]$$

- The corresponding K Matrix:

$$10^5 \begin{bmatrix} 0.0006 & 0.0033 & -0.0050 & 0.1409 & 0.0009 & 0 & 0.0015 & 0.0209 & -0.0183 & 0.0760 & 0.0053 & 0 \\ 0.00311 & 0.0637 & -0.0037 & -5.0920 & -1.0068 & 0 & 0.1662 & 0.5142 & -0.0061 & 5.5415 & -0.5095 & 0 \\ -0.0771 & -0.0249 & -0.0002 & -0.8712 & 5.4082 & 0 & -0.5736 & -0.1398 & -0.0006 & -0.4483 & 5.6859 & 0 \end{bmatrix}$$

- Chosen Disturbances:
  - For Height and Roll:
    - I chose a pulse generator that generates a spike with an amplitude of 100 every 100 seconds for 10% of the period.
  - For Pitch
    - I chose to keep the same disturbance that Dr. Brooks gave us in the Simulink model.

- My Simulink Model:

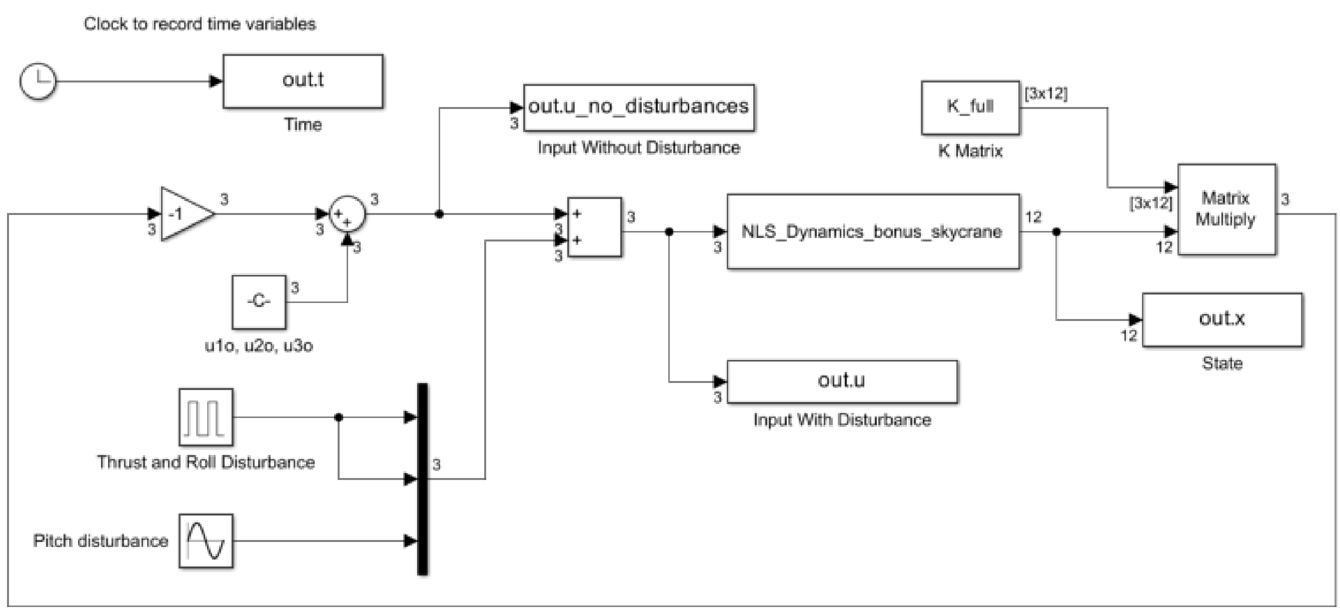

Figure 14: *Pole Placement Sim*

- Plots:

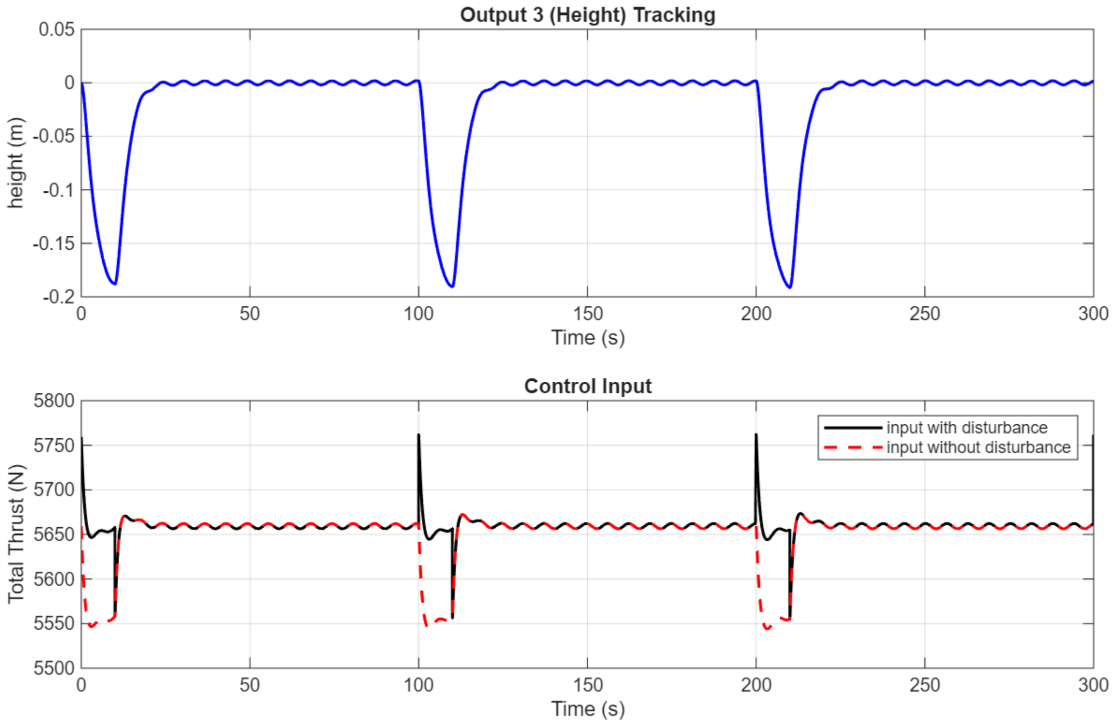

Figure 15: *$\tilde{y}_3$ (Height) and its reference vs time and $\tilde{u}_1$ (Total Thrust) vs time.*

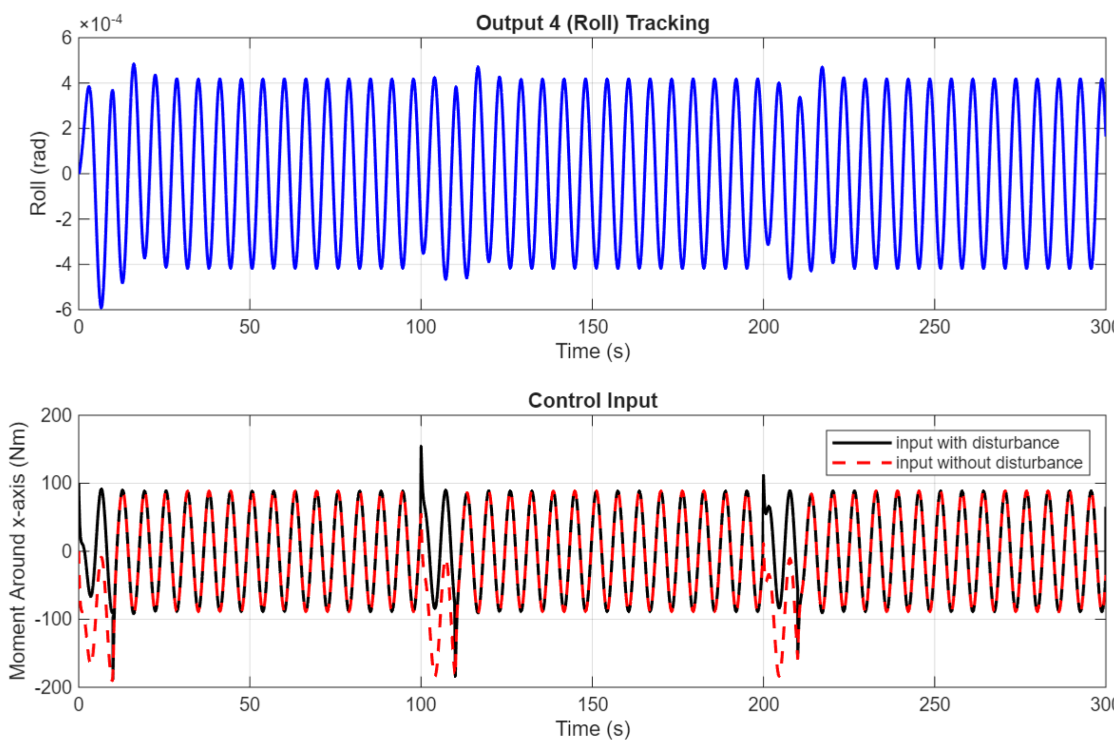

Figure 16: *$\tilde{y}_4$ (Roll) and its reference vs time and $\tilde{u}_2$ (Total Moment about x-axis) vs time.*

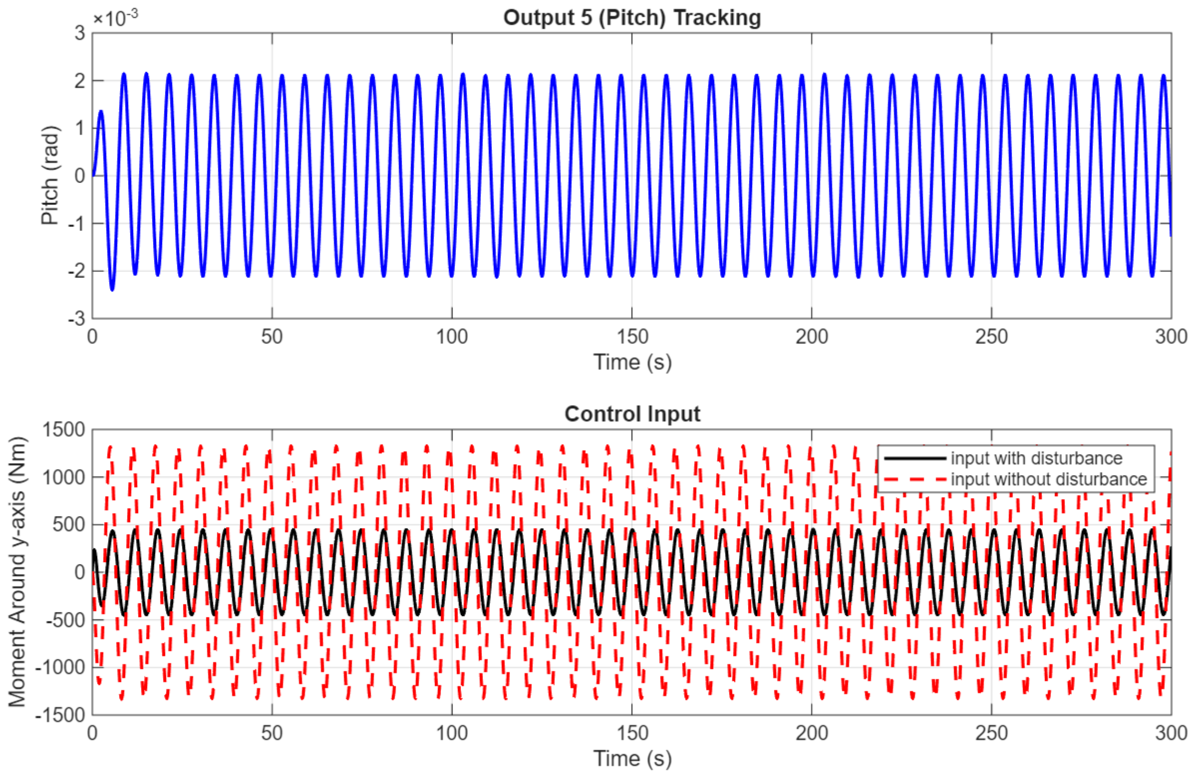

Figure 17: *$\tilde{y}_5$ (Pitch) and its reference vs time and $\tilde{u}_3$ (Total Moment about y-axis) vs time.*